Breast Cancer detection in mammograms through image processing and convolutional neural networks

1st Andrea Carolina Terraza Sequea, 2nd Darwin Eduardo Martinez Riaño
Computer Science and Artificial Intelligence Engineering, Universidad Sergio Arboleda, Bogotá, Colombia
Assistant Professor, Departamento de Estadística, Facultad de Ciencias, Universidad Nacional de Colombia, Bogotá, Colombia
andreats0729@gmail.com, demartinezr@unal.edu.co

***Abstract*—Breast cancer is a disease with high mortality in women, and its early detection is key to improving survival rates. This study aims to develop a computer-aided diagnosis (CAD) system based on convolutional neural networks (CNN) for classifying mammograms into normal, benign, and malignant categories. Using advanced image preprocessing techniques combined with deep learning strategies, a custom CNN model and a transfer learning model based on the VGG16 architecture achieved accuracies of 83.07% and 82.07%, respectively. Highlighting their ability to identify normal images with an AUC-ROC greater than 0.97. These results evidence that integration of image processing techniques with deep learning models can improve the accuracy of CAD systems in breast cancer detection.**

I. Introduction

Breast cancer remains a leading cause of mortality among women worldwide, with 2,296,849 new cases and 666,103 deaths reported annually [1]. Although it can affect women from puberty onward, its incidence peaks in adulthood [2]. According to the Global Cancer Observatory, early detection is critical to improving survival rates. Achieving a 99% success rate when it is diagnosed in initial stages (Stage I) [3]. A mammography image is used in the diagnosis procedure. This kind of image enables visualization of non-palpable or non-visible abnormalities, yet dense breast tissue can obscure microcalcifications and small masses [4], [5].

To address these challenges, Computer-Aided Detection (CAD) systems automate mammogram analysis, reducing diagnostic time and improving lesion detection, segmentation, and classification to support surgeons and radiologists in more accurate decision-making [6], [7], [8]. In this work, convolutional neural networks (CNNs) are investigated as the core of a CAD system, leveraging their capacity to learn discriminative image features and thereby enhance the accuracy of tumor and anomaly identification in mammograms [9].

II. Mammographic Image Classification for Cancer Detection

Mammographic image classification has been widely explored to improve early breast cancer detection. Preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) enhance local contrast and highlight critical patterns like microcalcifications and masses, enabling CNNs to learn more relevant features [10], [11].

Given the limited availability and diversity of annotated mammographic datasets, data augmentation become essential. It improves generalization and reduces overfitting in deep models [12], [13]. Operations like rotation, flipping, and translation expand training samples and are particularly valuable for deep CNNs that require large datasets to learn robust representations [9].

Pretrained networks such as VGG16, InceptionV3, ResNet50, and EfficientNet have been adapted to mammography by modifying their output layers and fine-tuning [16]. VGG16 achieved 88% accuracy for microcalcification detection [17]; InceptionV3 reached 99% in binary malignant vs normal classification [18]; ResNet50 obtained 88% in multiclass breast irregularity detection [19], and EfficientNet, balancing network depth and resolution, reported 75% accuracy [20].

However, most of these results focus on binary classification, which simplifies clinical decision-making but does not reflect the complexity of real world scenarios [18]. Distinguishing normal, benign, and malignant cases simultaneously remains more challenging yet clinically relevant. This study focuses on such a multiclass approach, leveraging advanced preprocessing, augmentation, and deep architectures to improve CAD system performance [21].

III. Data definition

This study uses a combined dataset from two public digital mammography collections:

- CBIS-DDSM (Curated Breast Imaging Subset of DDSM) [14]: includes 1644 cases (753 calcifications, 891 masses), labeled as benign or malignant, with region segmentations and bounding boxes. Image resolution is typically around 2000×4500 pixels.
- MIAS (Mammographic Image Analysis Society) [15]: provides 322 resized images (1024×1024 pixels), categorized as normal, benign, or malignant, with annotated anomaly types and locations.

Both datasets are widely used in breast cancer detection research [10]. They were merged to create a three-category dataset, which includes 3,608 mammograms: 1,995 benign, 1,406 malignant, and 207 normal. Normal images only came from MIAS, while of the over 10,000 mammograms in CBIS-DDSM, about one-third are adequately labeled for CAD use,

representing a significant constraint. Then, data availability remains a limiting factor.

IV. Methodology

This study tackles the classification of mammograms into three categories: normal, benign, and malignant using deep learning. Data from the CBIS-DDSM and MIAS datasets were first integrated and preprocessed. CNNs are implemented (pretrained and one customized), and their performance is evaluated with standard metrics (accuracy, AUC-ROC, confusion matrix), aiming to develop a CAD system to support a more accurate diagnosis of breast cancer.

A. Data Preprocessing

To ensure consistency for CNN input, all mammograms were first normalized (pixel intensities scaled to [0,1]) and resized to 224×224 pixels using bicubic interpolation [22]. To reduce noise and remove irrelevant artifacts, a Gaussian blur was applied, followed by thresholding to generate a binary mask isolating breast tissue. Next, a morphological operation (opening) was used to eliminate small artifacts, keeping only the largest connected component (breast region). Then, a contrast enhancement using CLAHE improves local contrast without noise amplification [10]; also, gamma correction adjusted the tonal range to highlight subtle tissue differences [23]. These transformations resulted in noise-reduced, contrast-enhanced, and standardized inputs for deep learning models.

B. Mammography classification process

Figure 1 presents the overall workflow, highlighting the required stages (i.e., preprocessing, data augmentation, model training) to final mammography classification into three classes. The main goal is to increase the model's ability to generalize and detect specific patterns in mammograms by using convolutional neural network architectures.

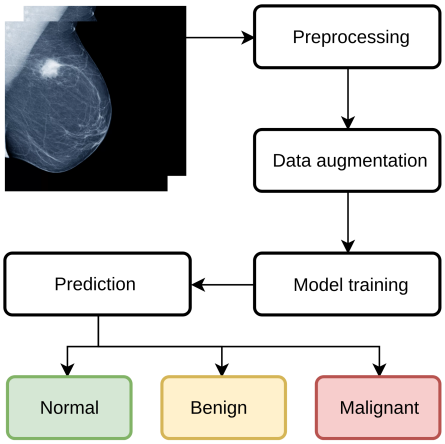

Fig. 1. Process applied for the classification of mammograms by deep learning. It starts from an original image, which undergoes preprocessing and data augmentation to improve the dataset. The images are divided into training and testing to train a model that classifies mammograms into normal, benign or malignant, supporting medical diagnosis.

A subject-wise partitioning strategy was employed to divide the combined mammography dataset into 70% training, 15% validation, and 15% test sets, ensuring that all images from any single patient (cranio-caudal CC and medio-lateral oblique MLO views) remained within the same subset to prevent information leakage. Data augmentation (random rotations of up to 20° and zoom adjustments of up to 20%) was applied exclusively to the training subset. This procedure expanded the training set to 5,995 images in total, with approximately 1,995 samples per class (normal, benign, malignant). Despite these augmentation efforts, the effective dataset size remains modest, which may limit the model's ability to generalize to truly novel, real-world mammograms.

C. Architectures

1) Pre-trained Architectures: EfficientNetB2, ResNet50, and VGG16 were selected based on their reported effectiveness in breast cancer classification tasks [17], [19], [20]. These models were used following a transfer learning approach. Initially, all convolutional layers were frozen to preserve the features learned during training on large-scale datasets such as ImageNet, particularly low-level and mid-level visual representations [24]. A custom classification head was then added to each architecture, consisting of a fully connected (dense) layer with 1024 neurons and ReLU activation, followed by a softmax output layer with 3 neurons for multiclass classification (normal, benign, malignant). In a second phase, fine-tuning was applied to improve model adaptation to mammographic data. Approximately 20–25% of the deepest convolutional layers in each architecture were unfrozen and retrained, while keeping earlier layers frozen to avoid overfitting [25].

2) Custom CNN Model: A custom convolutional neural network was developed for the three-class classification task. The architecture progressively increases the number of convolutional filters by layer, from 32 to 512, learning features from low to high levels. Also, it includes Max pooling layers for dimensionality reduction and computational efficiency. A fully connected layer of 1024 neurons with L_2 regularization (0.0001) and Dropout (0.3) enables robust high-level representation learning. ReLU activations were used in convolutional and dense layers to avoid vanishing gradients, and the Adam optimizer with a validation-tuned learning rate was employed for efficient convergence. This design, inspired by [26], [27], balances feature extraction and overfitting control while maintaining strong classification performance in medical imaging.

D. Models Validation and Robustness

To ensure the reliability of the results, each architecture was independently trained 20 times using the same training, validation, and test subjects. For each run, the best performing weights were selected based on the highest validation accuracy and then evaluated on the independent test set. The test accuracies obtained from all runs were used to compute the mean and standard deviation, thus quantifying model stability. This repeated evaluation mitigates the impact of randomness in weight initialization, providing a robust estimate of each model's generalization capability.

E. Performance metrics

To assess the models' performance in early breast cancer detection from mammograms, widely recognized metrics in medical imaging were employed: accuracy, area under the receiver operating characteristic curve (AUC-ROC), and the confusion matrix.

Accuracy provides a global measure of correctly classified samples, minimizing false positives and negatives that could delay or complicate diagnosis [28]. The AUC-ROC evaluates the discriminative ability of the models, even when class differences are subtle. Both macro and micro-averaging were used to capture class specific and overall performance [29].

Finally, the confusion matrix offers insight into misclassification patterns, particularly errors between benign and malignant cases, which directly impact clinical decision-making. This analysis guides further refinements in preprocessing and network design to improve sensitivity and specificity [30].

V. Results

The outcomes of the different implemented models are presented in this section. Each model was trained 20 times using the same data partitions, and the best performing instance for each architecture was selected based on the highest accuracy obtained on the test set.

Models were evaluated using accuracy, macro AUC-ROC, and the confusion matrix, which together provide a comprehensive view of their ability to classify mammograms into normal, benign, and malignant categories. Table I summarizes the best test accuracy and macro-AUC achieved by each model. These results shows the superior performance of the custom CNN followed by VGG16, as well as the challenges faced by ResNet50 and, in particular, EfficientNetB2.

TABLE I
COMPARISON OF FROZEN AND FINE-TUNED PRE-TRAINED MODELS AND THE CUSTOM CNN USING TEST ACCURACY, MACRO AUC, AND MEAN ± STANDARD DEVIATION OVER 20 RUNS.

Model	**Config.**	**Accuracy (best)**[a]	**Macro AUC**	**Mean ± Std**
VGG16	Frozen	82.07%	0.89	80.97% ± 0.0056
	Fine-tuned	81.29%	0.88	56.17% ± 0.2290
ResNet50	Frozen	75.50%	0.65	73.64% ± 0.0094
	Fine-tuned	77.39%	0.66	72.23% ± 0.0291
EfficientNetB2	Frozen	33.59%	0.61	33.39% ± 0.0004
	Fine-tuned	75.06%	0.66	60.09% ± 0.1923
Custom CNN	N/A	83.07%	0.90	81.27% ± 0.0054

[a]Best test accuracy among 20 runs.

A. Performance of the Best Model: Custom CNN

The custom CNN achieved the best overall performance with a test accuracy of 83.07%, slightly outperforming all pre-trained models. Its AUC-ROC scores were 0.98 (normal), 0.88 (benign), and 0.85 (malignant), indicating excellent sensitivity to normal images and good performance for the benign and malignant classes.

Confusion matrix shows that normal and malignant cases were mostly classified correctly, though 102 malignant samples were confused with benign. ROC curves confirms strong discrimination, especially for normal cases.

VI. Discussion

This study proposed a multiclass CNN-based CAD system for classifying mammograms into normal, benign, and malignant categories. Two public datasets (CBIS-DDSM and MIAS) were merged, preprocessed using CLAHE and artifact removal, and expanded with data augmentation to improve generalization. Both custom CNN and pre-trained architectures (VGG16, ResNet50, EfficientNetB2) were evaluated using frozen and fine-tuned configurations.

The custom CNN achieved the highest test accuracy (83.07%) and very low variability (0.0054) across 20 runs, demonstrating consistent and robust performance. VGG16 with fine-tuning yielded competitive accuracy (81.29%) but suffered high variance, while deeper models like ResNet50 and EfficientNetB2 improved with fine-tuning yet remained less effective, especially in malignant case detection. These findings reinforce that fine-tuning 20–25% of deep layers can help align pre-trained models to the mammographic domain, but model stability depends on architectural complexity and dataset fit.

Preprocessing significantly impacted performance. CLAHE effectively enhanced contrast and revealed subtle patterns such as microcalcifications, which are critical for early detection [10]. It is important to emphasize that preprocessing must be tailored to dataset specific noise and resolution characteristics.

Clinically, three class classification is highly relevant: normal images indicate routine monitoring, benign lesions may require observation or biopsy, and malignant findings demand immediate intervention. Such detailed categorization enhances diagnostic specificity and reduces unnecessary procedures.

Nevertheless, a persistent challenge across all models was the confusion between benign and malignant lesions, underscoring the complexity of subtle tissue variations. Future research should explore improved mass boundary delineation, texture descriptors, or integration of clinical metadata to improve this distinction.

One critical limitation is the exclusive use of standard mammographic views (CC and MLO), which do not fully visualize regions like the axilla and pectoral muscle [31]. This creates a clinical blind spot where metastatic or peripheral lesions may go undetected. In practice, radiologists supplement mammography with modalities like axillary ultrasound, MRI, or PET-CT to ensure complete coverage [32]. Therefore, future CAD systems must incorporate multiple views and multimodal data to minimize diagnostic gaps and improve real world applicability.

Dataset diversity also remains a barrier. Only a subset of CBIS-DDSM images were usable, and normal cases came solely from MIAS, limiting demographic variability. To enhance generalization, future work should prioritize datasets that include broader anatomical, ethnic, and geographic representation especially underrepresented populations and consider synthetic data augmentation.

To sum up, the custom CNN proved most reliable under current constraints, with fine-tuned transfer learning models showing potential but requiring careful adaptation. Addressing

data diversity, projection limitations, and multimodal integration will be useful to advance CAD systems for multiclass breast cancer detection from research tools to reliable clinical assistants.

VII. Conclusions

This work contributes to the advancement of CAD systems for early breast cancer detection. Findings on preprocessing strategies, data augmentation, and architecture adaptation provide a foundation for future research. Further exploration of advanced preprocessing methods, integration of clinical information (e.g., age, family history, breast density), and development of hybrid or specialized architectures are recommended. The ultimate goal is to develop CAD tools that approach the performance of expert radiologists, improving early detection, reducing unnecessary biopsies, and, ultimately, enhancing patient prognosis and life quality.

References

[1] International Agency for Research on Cancer, "Global cancer observatory," *GCO*, 2024.

[2] World Health Organization, "Breast cancer," *WHO*, 2024.

[3] National Institutes of Health, "Cancer stat facts: Female breast cancer," *NIH*, 2024. [Online]. Available: https://seer.cancer.gov/statfacts/html/breast.html

[4] National Institutes of Health, "Breast cancer screening (PDQ®)–Patient version," *NIH*, 2023.

[5] American Cancer Society, "American Cancer Society Guidelines for Breast Cancer Screening", Cancer.org, 2023. [Online]. Available: https://www.cancer.org/es/cancer/tipos/cancer-de-seno/pruebas-de-deteccion-y-deteccion-temprana-del-cancer-de-seno.html

[6] N. M. Hassan, S. Hamad, and K. Mahar, "Mammogram breast cancer CAD systems for mass detection and classification: a review," *Multimedia Tools and Applications*, vol. 81, no. 14, pp. 20043–20075, 2022.

[7] K. Loizidou, R. Elia, and C. Pitris, "Computer-aided breast cancer detection and classification in mammography: a comprehensive review," *Computers in Biology and Medicine*, vol. 153, p. 106554, 2023.

[8] A. Kumar and S. Sasikala, "Review on deep learning-based CAD systems for breast cancer diagnosis," *Technology in Cancer Research & Treatment*, vol. 22, 2023, Art. ID 15330338231177977. [Online]. Available: https://doi.org/10.1177/15330338231177977

[9] A. W. Salehi, S. Khan, G. Gupta, B. I. Alabduallah, A. Almjally, H. Alsolai, T. Siddiqui, and A. Mellit, "A study of CNN and transfer learning in medical imaging: advantages, challenges, future scope," *Sustainability*, vol. 15, no. 7, Art. ID 5930, 2023. [Online]. Available: https://www.mdpi.com/2071-1050/15/7/5930

[10] H. Avci and J. Karakaya, "A novel medical image enhancement algorithm for breast cancer detection on mammography images using machine learning," *Diagnostics*, vol. 13, no. 3, p. 348, 2023. [Online]. Available: https://www.mdpi.com/2075-4418/13/3/348

[11] D. Tribuana, Hazriani, and A. L. Arda, "Image preprocessing approaches toward better learning performance with CNN," *Jurnal RESTI*, vol. 8, no. 1, pp. 1–9, Jan. 2024.

[12] R. Sawyer-Lee, F. Gimenez, A. Hoogi, and D. Rubin, "Curated breast imaging subset of digital database for screening mammography (CBIS-DDSM)," The Cancer Imaging Archive, 2016. [Online]. Available: https://www.cancerimagingarchive.net/collection/cbis-ddsm/

[13] J. Suckling et al., "Mammographic image analysis society (MIAS) database v1.21," Apollo - University of Cambridge Repository, 2015. [Online]. Available: https://www.repository.cam.ac.uk/handle/1810/250394

[14] O. N. Oyelade and A. E. Ezugwu, "A deep learning model using data augmentation for detection of architectural distortion in whole and patches of images," *Biomedical Signal Processing and Control*, vol. 65, p. 102366, 2021.

[15] O. N. Oyelade and A. A. Sambo, "Data augmentation-aided convolutional neural network for detection of abnormalities in digital mammography," *SLU Journal of Science and Technology*, 2022.

[16] F. A. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, "Breast cancer histopathological image classification using convolutional neural networks," in *Proc. Int. Joint Conf. on Neural Networks (IJCNN)*, Vancouver, Canada, 2016, pp. 2560–2567.

[17] Z. Liu, J. Peng, X. Guo, S. Chen, and L. Liu, "Breast cancer classification method based on improved VGG16 using mammography images," *Journal of Radiation Research and Applied Sciences*, vol. 17, no. 2, p. 100885, 2024. [Online]. Available: https://doi.org/10.1016/j.jrras.2024.100885

[18] M. Rathi, A. A. Shah, S. Ali, E. N. Baro, A. W. Memon, and A. Inam, "Binary Inception V3 deep learning-based image classifier for the detection of breast cancer," in *Proc. Am. Inst. Phys. Conf. Ser.*, vol. 3125, Aug. 2024, Art. ID 060004. [Online]. Available: https://doi.org/10.1063/5.0214634

[19] J. Nalifabegam, C. Ganeshbabu, N. Askarali, A. Natarajan, and P. Maheshwari, "Cancer classification revolution: employing advanced deep CNNs for multi-class detection of breast irregularities," in *Proc. Int. Conf. on Smart Technologies, Communication and Robotics (STCR)*, 2023, pp. 1–4.

[20] S. Gengtian, B. Bing, and Z. Guoyou, "EfficientNet-based deep learning approach for breast cancer detection with mammography images," in *Proc. 8th Int. Conf. on Computer and Communication Systems (ICCCS)*, 2023.

[21] M. Prodan, E. Paraschiv, and A. Stanciu, "Applying deep learning methods for mammography analysis and breast cancer detection," *Applied Sciences*, vol. 13, no. 7, p. 4272, 2023.

[22] S. Aymaz, "A new framework for early diagnosis of breast cancer using mammography images," *Neural Computing and Applications*, vol. 36, pp. 1665–1680, 2024.

[23] M. K. Lim, W. Khairunizam, and W. A. Mustafa, "A computational investigation of breast tumour on mammogram based on pattern of grey scale distribution," *J. Biomimetics Biomaterials Biomed. Eng.*, vol. 43, pp. 67–73, 2019.

[24] F. Guzmán-Manekar, "Exploring layer freezing in transfer learning," Master's thesis, Dept. of Informatics, Univ. of Zurich, 2024. [Online]. Available: https://www.ifi.uzh.ch/dam/jcr:76acccb6-0dfd-4b5c-9df8-ee049133f9f9/fazla_guzman_manekar2024master_project.pdf

[25] A. Davila, J. Colan, and Y. Hasegawa, "Comparison of fine-tuning strategies for transfer learning in medical image classification," *arXiv preprint arXiv:2406.10050*, 2024. [Online]. Available: https://arxiv.org/abs/2406.10050

[26] A. H. Ahmed and M. A.-M. Salem, "Mammogram-based cancer detection using deep convolutional neural networks," in *Proc. 13th Int. Conf. Computer Engineering and Systems (ICCES)*, Cairo, Egypt, 2018, pp. 694–699. doi: 10.1109/ICCES.2018.8639224

[27] H.-C. Lu, E.-W. Loh, and S.-C. Huang, "The classification of mammogram using convolutional neural network with specific image preprocessing for breast cancer detection," in *Proc. 2nd Int. Conf. Artificial Intelligence and Big Data (ICAIBD)*, Chengdu, China, 2019, pp. 9–12. doi: 10.1109/ICAIBD.2019.8837000

[28] J. Ahmad, S. Akram, A. Jaffar, Z. Ali, S. M. Bhatti, A. Ahmad, and S. U. Rehman, "Deep learning empowered breast cancer diagnosis: advancements in detection and classification," *PloS One*, vol. 19, no. 7, p. e0304757, 2024. [Online]. Available: http://dx.doi.org/10.1371/journal.pone.0304757

[29] F. Pesapane, C. Trentin, F. Ferrari, G. Signorelli, P. Tantrige, M. Montesano, and E. Cassano, "Deep learning performance for detection and classification of microcalcifications on mammography," *Eur. Radiol. Exp.*, vol. 7, no. 1, p. 69, 2023.

[30] Y. Jiang, M. Yang, S. Wang, X. Li, and Y. Sun, "Emerging role of deep learning-based artificial intelligence in tumor pathology," *Cancer Commun.*, vol. 40, no. 4, pp. 154–166, 2020. *Cancer Communications*.

[31] American Cancer Society, "Limitations of mammograms," *Cancer.org*, 2019. [Online]. Available: https://www.cancer.org/es/cancer/tipos/cancer-de-seno/pruebas-de-deteccion-y-deteccion-temprana-del-cancer-de-seno/mamogramas/limitaciones-de-los-mamogramas.html

[32] National Cancer Institute, "Breast cancer screening (PDQ®)–Health professional version," *NIH*, 2024. [Online]. Available: https://www.cancer.gov/espanol/tipos/seno/pro/deteccion-seno-pdq